DATHEON INTERNSHIP PROJECT

TASK 2
Interactive Power BI Executive Sales & Profit Dashboard

Dashboard design, data modelling, DAX measures and executive reporting guide

SUBMITTED BY	INSTITUTION	DOMAIN
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Programme: Datheon Data Analytics & Power BI Internship | Batch 01
Dataset disclosure: synthetic demonstration dataset; final PBIX must be authored in Power BI Desktop.

1. Project Overview

The objective is to build a one-page executive dashboard that allows decision-makers to monitor revenue, profit, margin, order volume, average order value, product mix and regional performance. The dashboard is designed around the cleaned dataset produced in Task 1.

2. Data Preparation and Model

Import datasets/superstore_retail_clean.csv into Power BI Desktop and name the query Orders. Confirm date, decimal and whole-number types. Create a dedicated Date table spanning the minimum and maximum Order Date. Mark it as the model's date table and establish an active one-to-many relationship from Date[Date] to Orders[Order Date]. Use single-direction filtering from Date to Orders.

Table / field	Role
Orders	Retail transaction fact table containing customer, geography, product, sales, quantity, discount and profit.
Date	Calendar dimension for year, month, year-month and time intelligence.
Order Date relationship	Active Date[Date] → Orders[Order Date], one-to-many.
Optional inactive date	Ship Date can be added as an inactive relationship for shipping analysis using USERELATIONSHIP.

3. Core DAX Measures

The complete DAX library is supplied in task2_power_bi/dax/measures.dax. Key measures include Total Sales, Total Profit, Total Orders, Total Quantity, Average Order Value, Profit Margin %, Average Discount %, Sales YTD, Sales LY and Sales YoY %. Use DIVIDE() to avoid division-by-zero errors.

Measure	Definition
Total Sales	SUM(Orders[Sales])
Total Profit	SUM(Orders[Profit])
Total Orders	DISTINCTCOUNT(Orders[Order ID])
Average Order Value	Total Sales ÷ Total Orders
Profit Margin %	Total Profit ÷ Total Sales
Sales YoY %	(Current Sales – Sales LY) ÷ Sales LY

4. Dashboard Layout Specification

Visual	Fields / purpose
KPI cards	Total Sales; Total Profit; Profit Margin %; Total Orders; Average Order Value.
Monthly trend	Line chart: Date[Year-Month] with Total Sales and Total Profit.
Regional comparison	Bar chart: Region by Total Sales; Total Profit in tooltip.
Category performance	Clustered column: Category by Total Sales and Total Profit.
Top products	Bar chart: Product Name by Total Sales; Top N = 10.
Discount-profit analysis	Scatter: Discount vs Profit, Sales as bubble size.
Slicers	Year, Region, Segment, Category and Ship Mode.

5. Interactivity and User Experience

Enable cross-filtering between visuals and verify interactions in Power BI's Edit interactions pane. Add slicers for the principal business dimensions. A bookmark-based Reset Filters button is recommended. Use descriptive titles, consistent number formats, clear units and accessible contrast. Avoid excessive gridlines and unnecessary decorative visuals.

6. Validation and Quality Assurance

- Reconcile Total Sales and Total Profit with Task 1 Python summaries.
- Confirm the Date relationship is active and Date has unique dates.
- Confirm Profit Margin is calculated from aggregate profit divided by aggregate sales.
- Test each slicer and cross-filter interaction.
- Verify date filtering changes the monthly trend and KPI cards.
- Check Top N product ranking and currency/percentage formats.

7. Expected Deliverables and Current Build Status

This project package includes the cleaned data, DAX measure library, data-model instructions, dashboard wireframe specification, validation checklist and this report. The actual .pbix binary and final Power BI screenshots must be created in Power BI Desktop; they are not included or claimed as completed because Power BI Desktop is a separate authoring application.

8. Conclusion

The proposed dashboard translates transaction-level retail data into executive KPIs and interactive performance views. Combined with Task 1, it demonstrates a practical Python-to-Power-BI workflow and a reproducible portfolio project.